

AI for Coders & Software Engineers

The more I learn, the more I realize I don't know.

— *Albert Einstein* —

A useful mindset for this bootcamp

Mohammad Arshad • Instructor

21 years across AI, Data Science, Analytics & Enterprise Technology

ICEBREAKER

THEN

16-day ENEC deep dive

First cohort completed • January 2026

NOW

3.5 days accelerated learning

+ final half-day team capstone

PROMISE

Beginner or expert — everyone learns

Practical, hands-on, high-energy



EXPERIENCE BUILDING AND LECTURING AT



accenture

DELL Technologies

ماجد الفطيم
MAJID AL FUTTAIM

Microsoft



Software Engineer
2005 - 2010

absolutdata
-an lifogain company

Data Scientist
2010 - 2012



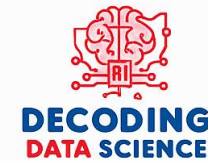
Data Scientist
2012 - 2014

accenture

Lead Data Scientist
2014 - 2021



Data Science Manager
2021 - 2022



Founder
2022 - 2025

DELL Technologies



Yale GAP

ICEBREAKER

What is one AI capability you would love to build or use confidently by Day 4?

Keep your answer — we will revisit it at the end of the programme.



<https://github.com/Decoding-Data-Science/enec2026>



Four projects. Four operational questions.

Four capstone ideas grounded in operational work.

01 Vendor proposals

Is the evidence enough to approve?

02 Workload manager

Where will capacity fall short?

03 OT asset cyber

Which change raises risk?

04 Supplier risk

What needs human review?



Start with the decision.

Then choose the data, architecture and technique.

NOT THIS

"Where can we put an LLM?"

INSTEAD

"Which decision or risk are we improving?"

Problem

Value

Data

Architecture

Technique

Application

Human
decision

The system prepares evidence. People remain accountable.

A strong idea needs a workable plan.

The last cohort was scored on both.

IDEA / 10

Relevant · useful · inventive

PLAN / 10

Clear · feasible · practical

TOTAL / 20

A good story is not enough. Show how you will test it.

CHOOSE THE TOOL

Same asset. Different questions.

Forecast and optimize first. Agents gather evidence within bounds.

WHAT HAPPENED?

SQL / analytics

WHAT NEXT?

Forecasting

IS THIS UNUSUAL?

Anomaly detection

WHAT KIND / HOW MUCH?

Classification / regression

WHAT IS THE BEST PLAN?

Optimization

WHAT DOES POLICY SAY?

Governed RAG

GATHER THE EVIDENCE?

Bounded agent

Build one asset end to end.

A-001 is synthetic. The architecture is the lesson.

01

**Sensor
evidence**

DAY 1

02

Data + ML

DAY 1

03

**Approved
knowledge**

DAY 2

04

Tools + agent

DAY 3

05

**Human
decision**

DAY 4

SENSORS → CLEAN DATA → PREDICT → RETRIEVE → ACT → REVIEW

One working pattern first. Scale only after validation.

YOUR TURN

By Day 4, build one decision aid.

Start with your job. Design backward from the human decision.

1 What problem?

2 Whose decision?

3 What value or risk?

4 Which data and documents?

5 Which technique?

6 What evidence and test?

7 Where does a human approve?

Build something useful. Keep the person in control.

Day 1

Databricks, Enterprise Data & Machine Learning

From a table of sensor readings to an honest anomaly flag.

- Four-day applied enterprise AI programme
- One synthetic scenario: Asset A-001, a cooling-water pump
- Hands-on: Databricks, SQL, Python, one honest anomaly classifier
- Today builds the data and ML foundation for RAG and agents

One Asset. Four Days. One Application.

The capability journey - each day adds one layer to the same A-001 story.



THE FIVE QUESTIONS WE WILL REPEAT

1. What do we know from the data?
2. What can we predict?
3. What does the organisation know?
4. What should the AI application do with it?
5. Where must a human remain responsible?

TEACHING RHYTHM

WHY - the enterprise problem first
WHAT - the new capability, smallest working version
HOW - hands-on in the notebook
SO WHAT - what it proves, and what it does not

The technology changes every day. The A-001 story stays the same.

Meet A-001: Our Single Enterprise Story

One synthetic cooling-water pump carries every exercise in the programme.

STRUCTURED EVIDENCE - TABLES

- Asset master and system context
- One-minute and hourly sensor readings
- Asset health scores
- Work orders, inspections, maintenance history
- Projects, risks and document links
- Agent runs, tool calls, human approvals

UNSTRUCTURED EVIDENCE - 9 DOCUMENTS

- 01 Reliability & Escalation Policy **APPROVED**
- 02 Operating & Usage Guide **APPROVED**
- 03 Inspection Procedure V1 **SUPERSEDED**
- 04 Inspection & PM Procedure V2 **APPROVED**
- 05 Inspection Procedure V3D **DRAFT**
- 06 Troubleshooting & Diagnostic Guide **APPROVED**
- 07 Condition Monitoring Report **APPROVED**
- 08 Inspection Work Order **OPEN**
- 09 Field Condition Report **APPROVED**

A database gives structured facts. Documents give organisational knowledge. Neither replaces the other.

The Enterprise Data Map

What we will use - one governed schema: workspace.nuclear_enterprise_360

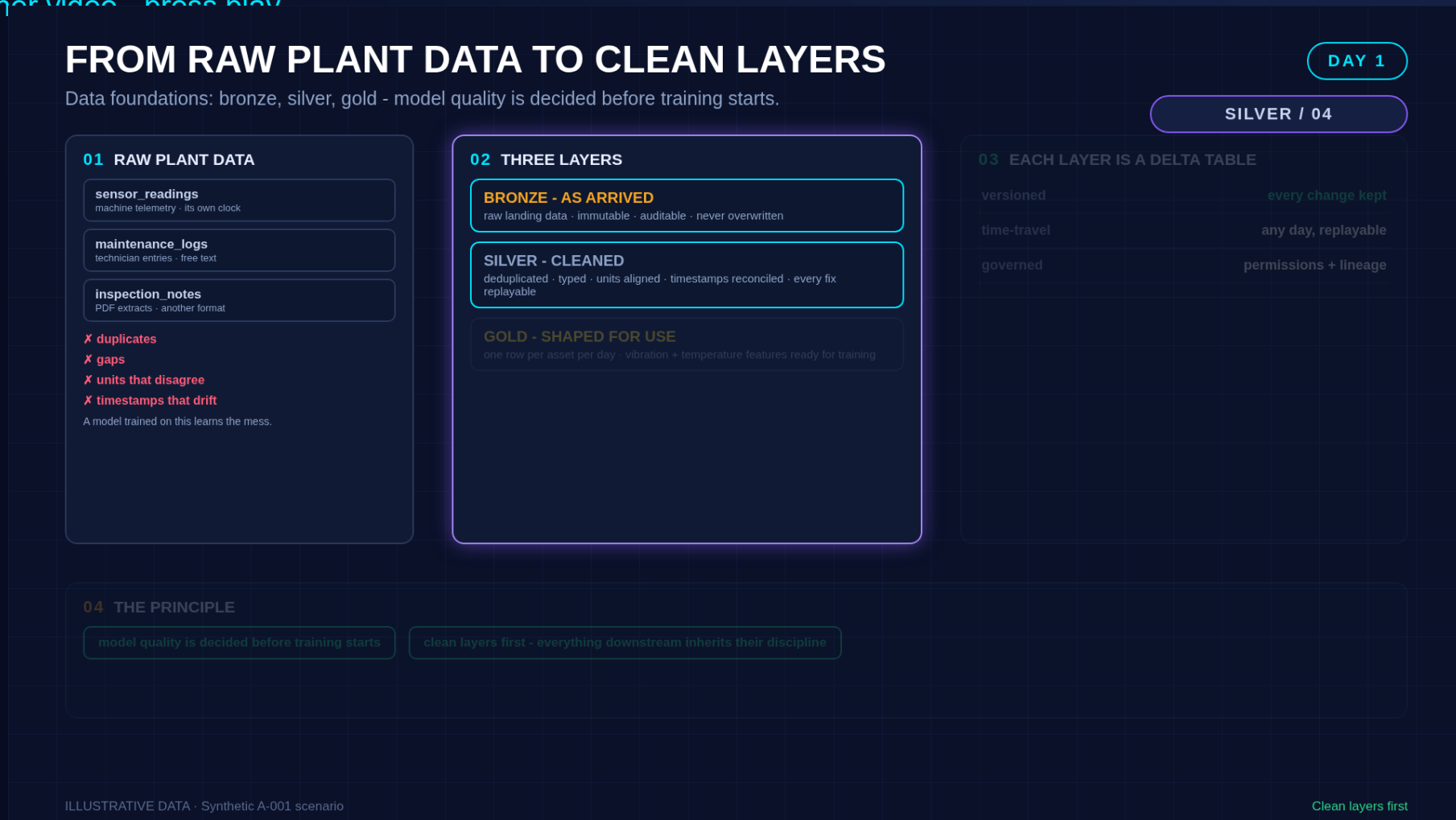
Object	Approx. rows	Teaching use
assets	128	Asset master
sensor_readings_a001_1min	525,600	Raw one-minute A-001 sensor history
a001_sensor_hourly	8,760	Hourly A-001 rollup
sensor_readings	1,820 (A-001)	Primary Day 1 ML dataset - carries the anomaly labels
asset_health_scores	11,520	Condition deterioration
work_orders / inspections / maintenance	1,500 / 600 / 900	Operational response + evidence
risks + projects	423 + 29	Enterprise context
documents + a001_source_documents	42 + 9	RAG corpus and register
agent_runs / tool_calls / approvals	50 / 90 / 35	Audit trail for Days 3-4

Views used repeatedly: asset_360 · sensor_readings · a001_event_timeline · asset_project_map · a001_document_evidence

Know the grain of every table before you query it: one asset, one hour, one event, one document.

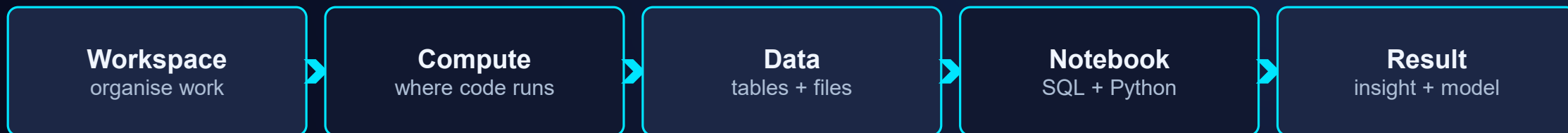
Watch: From Raw Plant Data to Clean Layers

Narrated explainer video - press play



Databricks: A Simple Mental Model

Five words you can navigate by - workspace, compute, data, notebook, result.



THE DATA HIERARCHY

catalog workspace

→ **schema** nuclear_enterprise_360

→ **tables / views** asset_360, a001_sensor_hourly, ...

→ **Volumes** governed file storage for the .db and PDFs

KEEP PLATFORM DETAIL IN ITS PLACE

- We use one small corner of a large platform.
- If setup fails live, switch to the pre-staged environment and keep teaching.
- The learning objective is enterprise AI, not platform administration.

If you can say where code runs and where data lives, you can navigate Databricks.

From a SQLite File to Governed Delta Tables

One setup notebook does the heavy lifting.



VALIDATION QUERIES

```
-- Find A-001
SELECT *
FROM workspace.nuclear_enterprise_360.asset_360
WHERE asset_id = 'A-001';

-- A-001 readings - expect 1,820
SELECT COUNT(*) FROM
workspace.nuclear_enterprise_360.sensor_readings;
```

READY WHEN

- ✓ Setup notebook runs clean
- ✓ asset_360 returns A-001
- ✓ sensor_readings: 1,820 A-001 rows
- ✓ Day 2 documents discoverable
- ✓ Day 3 skill files discoverable

Do not explain every line of setup code - explain the purpose of each section, then move on.

SQL Asks. Python Works.

Same dataset, two complementary ways to explore A-001.

SQL - ASK THE DATA

- Select rows and columns fast
- Filter, aggregate, group and join
- Quick validation of data quality
- Readable by analysts and auditors
- Every Day 3-4 tool starts as a bounded SQL question

PYTHON - WORK WITH THE DATA

- Reshape, aggregate and visualise
- Feature engineering for ML
- Train and evaluate models
- Glue between tables, models and files
- Notebooks mix both in one place

Rule of thumb: ask in SQL first; move to Python when the question needs reshaping, plotting or modelling.

While exploring ask: which columns are observations vs derived? what is the grain? what does a NULL mean here - missing, not applicable, or not yet recorded?

EDA: Three Ways to Look at A-001

Understand the data before any model sees it.

1 · UNIVARIATE - ONE VARIABLE

- What is typical? What is rare?
- Vibration: mean ≈ 1.36 mm/s
- Median ≈ 1.18 · range 0.88 - 5.75
- Outliers, skew, missing values
- `describe()` · `value_counts()` · `histogram`

2 · OVER TIME - SAME SENSOR

- A histogram shows distribution
- A time plot shows WHEN
- Sudden spikes, long shifts, drift
- Repeated patterns, missing periods
- Visual inspection → better ML questions

3 · BIVARIATE - TWO VARIABLES

- Vibration vs anomaly flag
- Normal max 4.977 / abnormal min 5.036
- A very clean separation
- Vibration vs temperature: $\text{corr} \approx 0.93$
- See separation before modelling

Never train on data you have not plotted first.

Watch: Look Before You Train - EDA on Asset 360

Narrated explainer video - press play

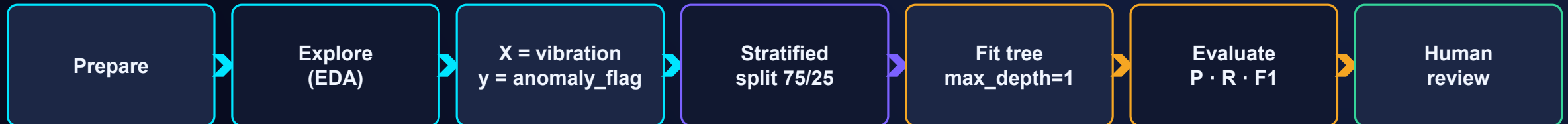


ML Starts with a Useful Question

Not with an algorithm.

THE QUESTION

"Can A-001 vibration readings classify a sensor reading as NORMAL or ABNORMAL?"



WHY ONE FEATURE ON PURPOSE

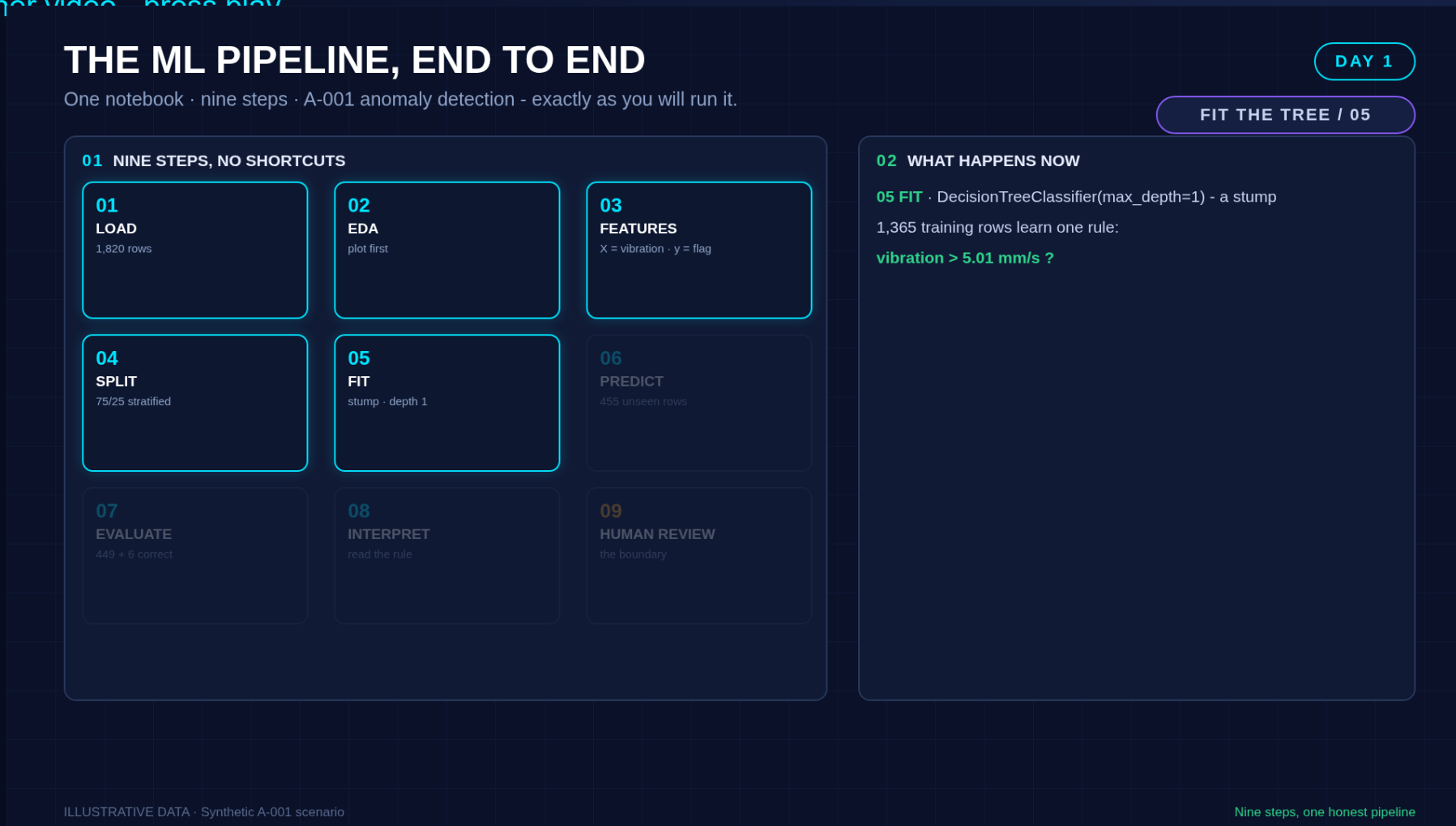
One input and one learned split ($\text{max_depth} = 1$) teach the complete ML lifecycle - prepare, explore, split, train, predict, evaluate - without hiding it behind model complexity. Multi-sensor models are a later upgrade, not a mystery.

Boundary: Abnormal $\neq$ pump failure. The label only says the reading looks unusual - prediction must match the label we actually have.

HANDS-ON Notebook: A001_Beginner_Anomaly_Detection_Databricks

Watch: The ML Pipeline, End to End

Narrated explainer video — press play



How a Decision Tree Flags A-001 Anomalies

One feature, one learned split - pick a reading and watch the path

Synthetic training simulation
Illustrative data - not plant output

THE SIMPLEST REAL CLASSIFIER - ONE FEATURE, ONE SPLIT (SYNTHETIC)



A-001 vibration in training data: mean $\approx$ 1.36 mm/s · median $\approx$ 1.18 · range 0.88 - 5.75 · normal max 4.977 vs abnormal min 5.036 - a clean separation

PICK A READING - WATCH THE PATH

Reading A: 4.8 mm/s

Reading B: 5.6 mm/s

Reading C: 7.1 mm/s

THE PATH THE TREE TAKES

Pick a reading above.

ON 455 HELD-OUT TEST ROWS

449 correct normal · 6 correct abnormal · 0 false alarms · 0 missed

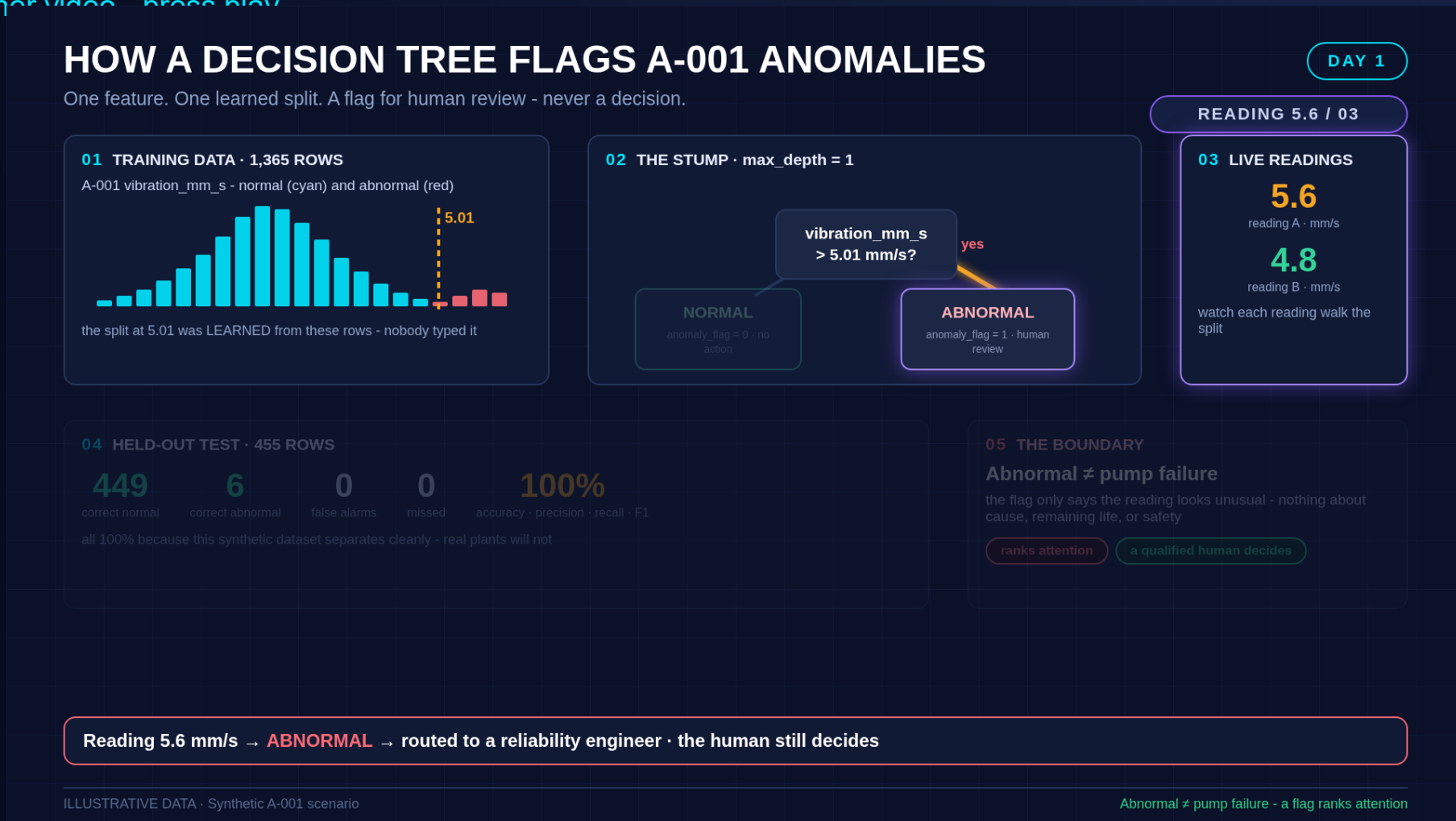
Accuracy, precision, recall, F1: all 100% - *because this synthetic dataset is deliberately easy*. Real plants will not score 100%.

ABNORMAL $\neq$ PUMP FAILURE

The flag only says the reading looks unusual. It ranks human attention - it never authorises maintenance, and it says nothing about remaining life or safety.

Watch: How a Decision Tree Flags A-001 Anomalies

Narrated explainer video - press play



Accuracy Is Not Enough

When anomalies are rare, one percentage hides the mistakes that matter.

THE CLASS-IMBALANCE TRAP

- 1,797 Normal · 23 Abnormal (1.26%)
- A model that always says NORMAL
- scores 98.7% accuracy -
- and catches nothing.
- Accuracy is necessary. Not sufficient.

THE QUESTIONS THAT MATTER

- Did we catch the abnormal cases? → Recall
- When we say abnormal, how often right? → Precision
- How is each class doing? → Confusion matrix
- Balance of both → F1 · balanced accuracy

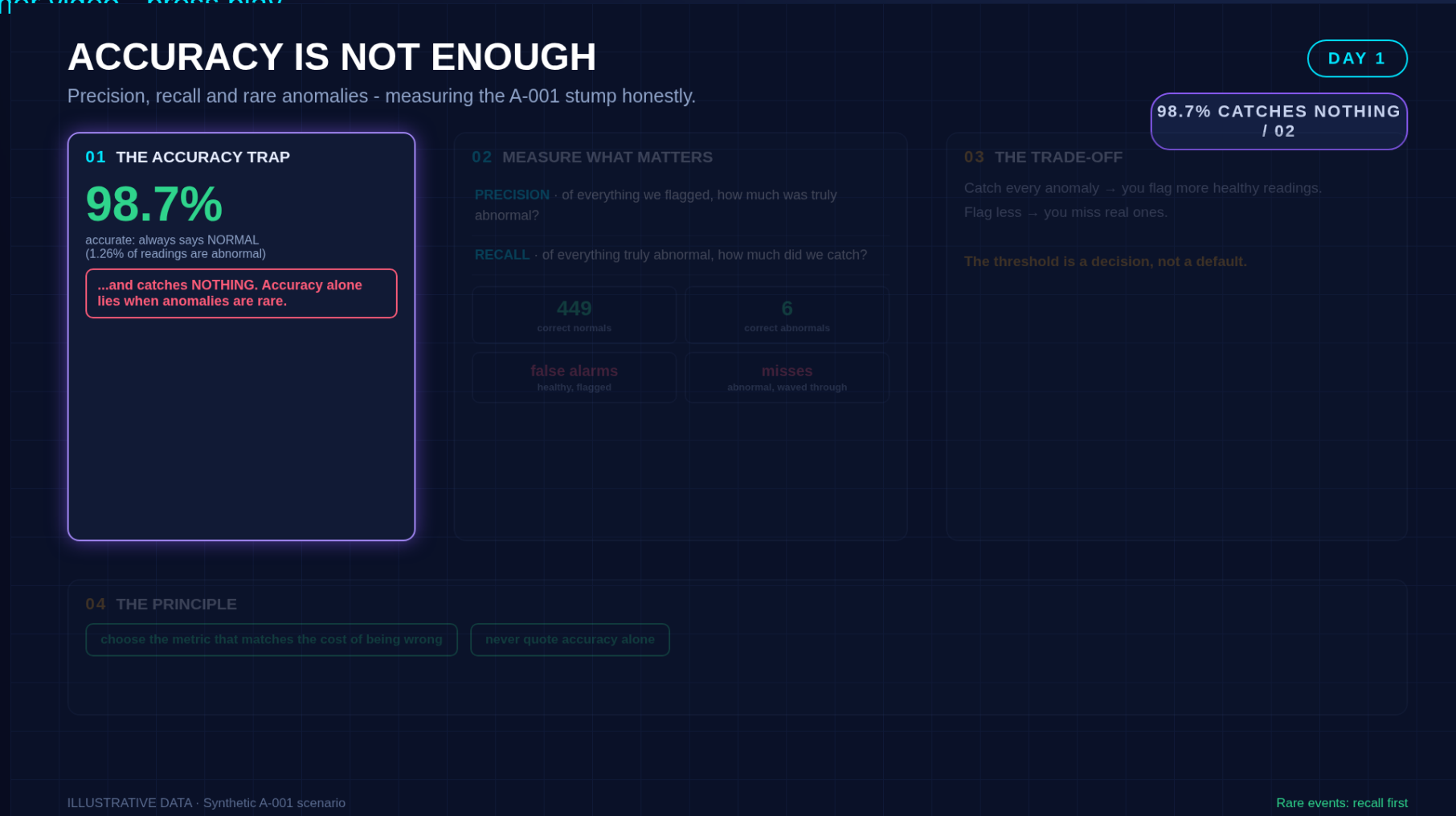
This **synthetic test set**: 449 correct normal · 6 correct abnormal · 0 false alarms · 0 missed - all metrics 100%. Real plants will not score 100%.

Would you authorise maintenance from this flag alone? Why not?

A flag is **evidence** - one input to a decision made by qualified people, with procedures, inspections and context the model has never seen. That missing context is not a failure of the model. It is the design of the enterprise - and it is exactly where Day 2 begins.

Watch: Accuracy Is Not Enough - Precision and Recall

Narrated explainer video - press play



Hands-on Mission: Prove the Loop

Everyone leaves Day 1 with a working environment and one honest anomaly classifier.

01 Open Databricks + attach compute

Workspace opens, notebook attached

02 Run setup and validate V2.2

asset_360 returns A-001

03 Inspect sensor_readings

1,820 A-001 readings confirmed

04 EDA: histogram + time plot

Distribution, and the clean separation

05 Train the one-feature tree

vibration_mm_s → anomaly_flag, max_depth=1

06 Record precision/recall + two sentences

What the flag can tell us - and cannot

If live setup fails: switch to the pre-staged environment and protect the learning objective.

Is a Flag Enough?

An anomaly flag without organisational context is not a decision.

WHAT DAY 1 GAVE US

- A working, governed data environment
- Explored A-001 with SQL and Python
- A trained, honestly scored anomaly classifier
- A clear sense of the model's limits

WHAT IS STILL MISSING

- What does the approved procedure require?
- What do reports and work orders already say?
- Which document version is authoritative?
- Who is allowed to act - and on what evidence?

DAY 2 Tomorrow: Generative AI + RAG - connecting the model to trusted A-001 knowledge.

Extra Slides for Projects

Saving time is only the start.

Ask what it costs when the decision goes wrong.

TIME SAVED

Useful, but only one measure

FAILURES

Catch signals earlier

OUTAGES

Protect availability

DELAYS

Avoid slow decisions

REWORK

Find gaps sooner

Which risk or decision would your project improve?

Build small. Test against reality.

A pilot should earn its claim with a baseline.

01

BASELINE

Current process

02

PILOT

Real scenarios

03

COMPARE

Quality + time

04

REVIEW

Human judgment

PLC / SCADA · HISTORIAN · WORK ORDERS · ERP · APPROVED DOCUMENTS

An agent gathers evidence. A person owns the decision.